8.3 dbplyr: Databases

A database stores tables for efficient retrieval. dplyr specifies the manipulation, dbplyr translates it into SQL, and collect() returns only the requested rows.

1. What is a database?

A **database** is an organized collection of data designed for efficient storage, retrieval, and management.

In R, analysis usually begins with a data frame stored in RAM. That representation is appropriate for a table such as `flights`. When the data are very large, when several users query them concurrently, or when they already reside in an external database, loading the entire collection into R may be inefficient or impossible.

A **relational database** organizes data into multiple tables that are connected by common identifiers (keys). A soccer database, for example, may contain the following tables.

Table	Information
Country	Country names and identifiers
League	Soccer leagues
Match	Match results, dates, and scores
Team	Team names and identifiers
Player	Player information

Unlike an ordinary data frame, these tables are stored together and can be queried without loading the entire database into memory.

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2. What is SQLite?

SQLite is a lightweight relational database management system (DBMS).

- `soccer.sqlite` is a database file that contains the soccer data.
- SQLite is the software that reads, manages, and queries that file.
- A single `.sqlite` file can contain multiple tables, indexes, and other database objects.

Other database management systems include MySQL, MariaDB, PostgreSQL, Microsoft SQL Server, and Google BigQuery. The relational model is shared; the software that manages the data differs.

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3. Understanding the R packages

Each component in the workflow has a distinct role.

Component	Role
R / RStudio	The environment in which R code is written and executed
dplyr	Provides `filter()`, `select()`, `mutate()`, and related verbs
dbplyr	Translates dplyr operations into SQL
SQL	The language used to query a relational database
DBI	Defines a standard interface between R and database systems
RSQLite	Implements the DBI interface for SQLite
SQLite	Executes SQL and manages the stored data
soccer.sqlite	The file that contains the database

`dbplyr`, `DBI`, and `RSQLite` are not attached by `library(tidyverse)`. They must be installed and loaded explicitly.

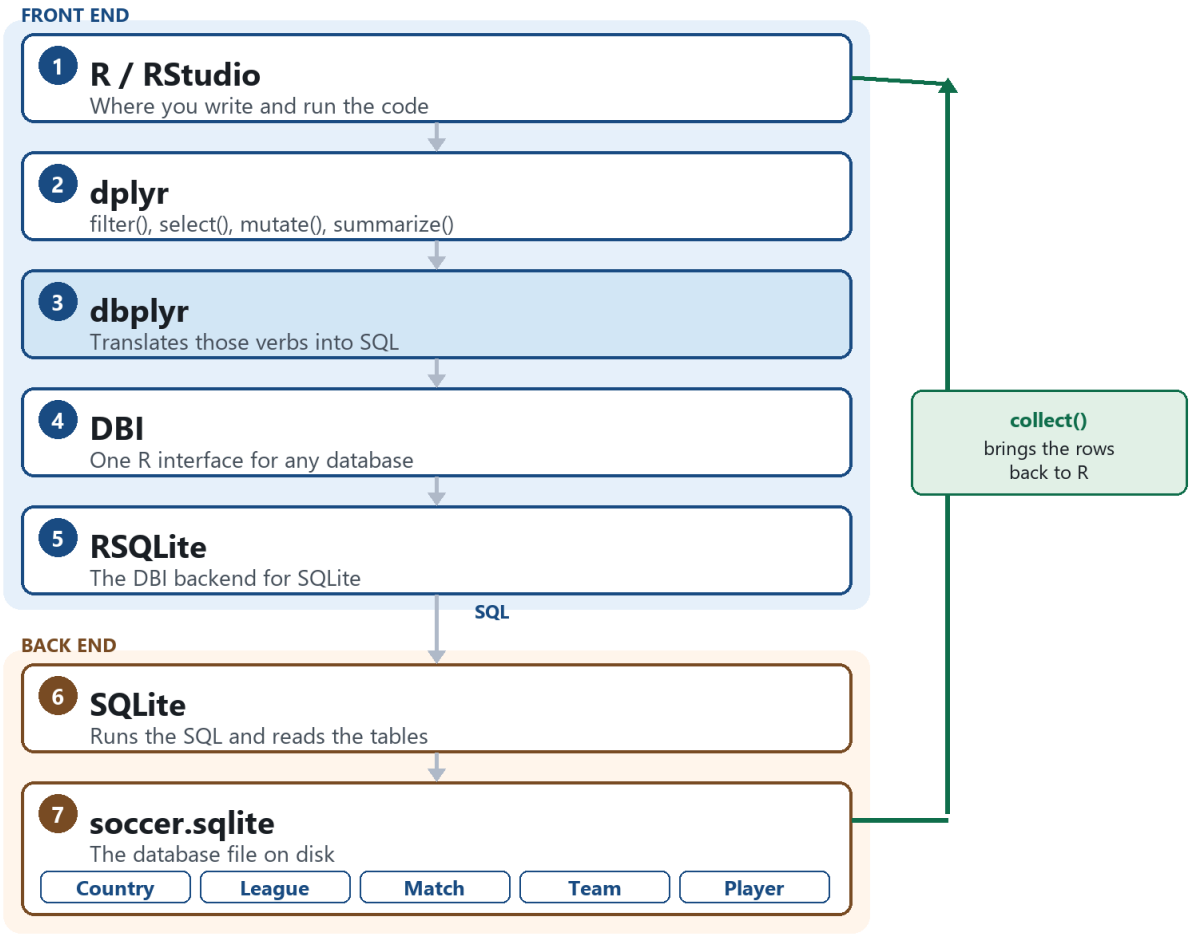
SQL is the language generated by `dbplyr` and executed by SQLite. It is not a separate software layer. DBI and RSQLite together establish the connection and submit the query.

Other packages that implement database connections include `RMariaDB`, `RPostgres`, `odbc`, and `bigrquery`. They allow R to communicate with other database systems. DBI remains the common interface, and dbplyr can still translate the dplyr code. Only the backend package changes.

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4. From the front end to the back end

The figure orders the components from the code written in R to the file on disk. The downward arrows are the query. The green path is the return of the result.



The front end is the R session. The back end is the database engine and the file stored on disk.

- **R / RStudio.** The environment in which the analysis is written and executed.
- **dplyr.** Specifies the manipulation through `filter()`, `select()`, `group_by()`, and `summarize()`.
- **dbplyr.** Translates those dplyr operations into SQL. dplyr does not communicate with the database file directly.
- **DBI.** Provides a standard set of functions for connecting, listing tables, and submitting queries. The same calls apply to SQLite and to other database systems.
- **RSQLite.** The SQLite implementation of DBI. It opens the `.sqlite` file.
- **SQLite.** Executes the SQL and processes the records.
- **soccer.sqlite.** The physical database file. `Match`, `Team`, `Player`, `League`, and `Country` are stored in that file, not in R's memory.

`collect()` is the return path. SQLite has performed the computation. Only the requested rows are transferred into R.

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5. Example: working with the soccer database

The database file in this example is `./data/soccer/soccer.sqlite`, relative to the project directory.

The pipe `|>` passes the current table into the next function. The same pipelines are often written with `%>%`.

Step 1. Load the required packages.

```
library(DBI)
library(RSQLite)
library(dplyr)
library(dbplyr)
```

Step 2. Connect to the database. `dbConnect()` opens the file. The object `con` is the connection, not a data frame.

```
con <- dbConnect(
  SQLite(),
  dbname = "./data/soccer/soccer.sqlite"
)
```

Step 3. Identify the available tables.

```
dbListTables(con)
```

Step 4. Create a reference to the Match table. `tbl()` does not download `Match`. `Match_db` is a lazy reference: a description of a query, not a data frame in RAM.

```
Match_db <- tbl(con, "Match")
```

Step 5. Manipulate the table with dplyr. The verbs are recorded. SQLite has not yet executed them.

```
result <- Match_db |>
  filter(season == "2010/2011") |>
  select(date, home_team_goal, away_team_goal)
```

Step 6. Examine the SQL translation. `show_query()` prints the SQL that dbplyr will submit.

```
show_query(result)
```

The generated SQL has this form:

```
SELECT date, home_team_goal, away_team_goal
FROM Match
WHERE season = '2010/2011'
```

Step 7. Retrieve the result into R. `collect()` executes the query and returns a tibble.

```
matches <- result |>
  collect()
```

`matches` is then a local tibble stored in R's memory. `Match_db` and `result` remain query descriptions.

Step 8. Close the connection.

```
dbDisconnect(con)
```

The connection should be closed explicitly. Otherwise R closes it when the session ends, and a file-backed database may remain locked until then.

The same sequence applies when `soccer.sqlite` is not available. `dbConnect(SQLite(), ":memory:")` creates a database that exists only for the current session, and `copy_to()` writes a local data frame, such as `nycflights13::flights`, into that database. `tbl()`, the dplyr verbs, `show_query()`, and `collect()` are unchanged.

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6. Why `collect()` is required

Before `collect()`. The query remains associated with the database. Supported operations are executed by SQLite when the query runs. Printing a lazy table displays a preview, commonly the first rows, rather than the full table.

After `collect()`. The selected rows form a tibble in RAM. Subsequent operations run locally in R. ggplot2 requires a local data frame, so a plot is constructed only after `collect()`.

Not every tidyverse verb has an SQL translation. `pivot_longer()`, `separate()`, and most tidyr operations require a local data frame. `filter()`, `select()`, `mutate()`, `arrange()`, `group_by()`, `summarize()`, and the join functions generally translate.

The purpose of dbplyr is to filter, join, and summarize large tables inside the database, and to transfer only the required result.

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7. Key takeaway

The path of a query is:

R, then dplyr, then dbplyr, then DBI and RSQLite, then SQLite, then `soccer.sqlite`.

- R is the environment in which the analysis is conducted.
- dplyr expresses the data manipulation.
- dbplyr converts that manipulation into SQL.
- DBI and RSQLite provide the connection and submit the query.

- SQLite performs the computation.
- `soccer.sqlite` stores the data.
- `collect()` returns the result to R.

The analyst writes ordinary dplyr code and does not load the entire database into memory.

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8. Practice

1. Name the seven components from R to `soccer.sqlite`. Which component is a language rather than a program?
2. Connect to `soccer.sqlite`. From `Match`, retain the 2010/2011 season and the columns `date`, `home_team_goal`, and `away_team_goal`. Call `show_query()`, then `collect()`. State how `matches` differs from `Match_db`.
3. Disconnect. If `soccer.sqlite` is not in the project, repeat the procedure with `nycflights13::flights` copied into an in-memory SQLite database, and restrict the rows to `month == 1`.